Snowflake Project - Data Engineering & SQL Solution

Date	29 October 2026
Team ID	Individual Project - Ujjawal Rathod
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	5 Marks

Solution summary

The proposed Snowflake implementation prepares the two EV CSV sources for consistent analysis, uses a layered data model, and serves curated records to KPI queries and visualization tools. The names below are recommended project names, not claims that the objects have already been created in a live Snowflake account.

SQL examples use a canonical curated schema. Map the source CSV headers to these aliases in the staging view before running: MODEL_YEAR, MARKET_REGION, VEHICLE_CATEGORY, MANUFACTURER, BATTERY_TYPE, DRIVING_RANGE_KM, ENERGY_CONSUMPTION_WH_KM, CUSTOMER_RATING, MANUFACTURING_COUNTRY, SALES_VOLUME. The supplied handoff did not include the raw CSV files or a live Snowflake connection, so numeric SQL outputs are intentionally not invented.

Project repository: https://github.com/ujjawalrathod404/Electric-Vehicle-market-performace-insights

Field	Details
Project / Repository Link	https://github.com/ujjawalrathod404/Electric-Vehicle-market-performace-insights
Data Source(s)	ev_market_train.csv (~200,000 rows) + ev_market_test.csv (~100,000 rows); about 300,000 combined records per prior handoff (verify in Snowflake).
Database	Proposed: EV_MARKET_DB
Warehouse	Proposed: EV_MARKET_WH (X-Small, auto-suspend enabled; use a size appropriate to the class account and suspend when idle).
Tables / Views Created	Proposed layers: RAW.EV_MARKET_RAW -> STAGING.EV_MARKET_STG -> CURATED.EV_MARKET_CURATED; MART views for category sales, year trend, manufacturer share, battery mix, range, ratings and region sales.
Key Transformations / Calculations	Null/duplicate profiling; text standardization; safe numeric parsing; year/rating/range validity flags; sales share, average range and average rating.
KPIs created	Sales volume and mix, yearly trend, Top 10 market share, battery mix, average range, energy use, rating, and country/region sales.
Key Insights	Prior summary: scooters lead category sales; about 200 manufacturers, top share ~0.6%; NMC/LFP dominate battery mix; category range spans about 105-550 km; ratings about 4.08-4.09/5. Verify in live outputs.

Recommended setup SQL

```
CREATE DATABASE IF NOT EXISTS EV_MARKET_DB;
CREATE SCHEMA IF NOT EXISTS EV_MARKET_DB.RAW;
CREATE SCHEMA IF NOT EXISTS EV_MARKET_DB.STAGING;
CREATE SCHEMA IF NOT EXISTS EV_MARKET_DB.CURATED;
CREATE SCHEMA IF NOT EXISTS EV_MARKET_DB.MART;

CREATE WAREHOUSE IF NOT EXISTS EV_MARKET_WH
  WAREHOUSE_SIZE = 'XSMALL'
  AUTO_SUSPEND = 60
  AUTO_RESUME = TRUE
  INITIALLY_SUSPENDED = TRUE;
```

Example curated KPI view

```
CREATE OR REPLACE VIEW EV_MARKET_DB.MART.VW_CATEGORY_SALES AS
SELECT VEHICLE_CATEGORY,
      SUM(SALES_VOLUME) AS TOTAL_SALES_VOLUME,
      COUNT(*) AS RECORD_COUNT
FROM EV_MARKET_DB.CURATED.EV_MARKET_CURATED
GROUP BY VEHICLE_CATEGORY;
```

Snowflake quality checklist

#	Criteria	Status
1	Required data is loaded and accessible in Snowflake	Pending live verification
2	Data is cleaned, transformed, and prepared appropriately	SQL workflow specified; execute and validate
3	Required tables/views and relationships are implemented	Proposed names; confirm actual objects
4	SQL logic, calculations, and KPIs are implemented	Queries documented; compare outputs to workbook
5	Outputs are validated against requirements	Capture actual result grids and reconciliation
6	Solution is organized and documented	Code and layer plan provided